$$\frac{2(z+2)+3}{(z+2)}dz = dx$$

$$\left(2+\frac{3}{z+2}\right)dz = dx$$

Integrating both sides, we get

$$2z+3\log|z+2| = x+C$$

That is, $2(x-y)+3\log|x-y+2| = x+C$.

Example 10.16

Solve : $\frac{dy}{dx} = (3x+y+4)^2$.

Solution

To solve the given differential equation, we make the substitution $3x+y+4=z$.

Differentiating with respect to x, we get $\frac{dy}{dx} = \frac{dz}{dx} - 3$. So the given differential equation becomes $\frac{dz}{dx} = z^2+3$.

In this equation variables are separable. So, separating the variables and integrating, we get the general solution of the given differential equation as $\frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{3x+y+4}{\sqrt{3}}\right) = x+C$.

EXERCISE 10.5

1. If F is the constant force generated by the motor of an automobile of mass M, its velocity V is given by $M\frac{dV}{dt} = F-kV$, where k is a constant. Express V in terms of t given that $V=0$ when $t=0$.
2. The velocity v, of a parachute falling vertically satisfies the equation $v\frac{dv}{dx} = g\left(1-\frac{v^2}{k^2}\right)$, where g and k are constants. If v and x are both initially zero, find v in terms of x.
3. Find the equation of the curve whose slope is $\frac{y-1}{x^2+x}$ and which passes through the point $(1,0)$.
4. Solve the following differential equations:

   (i) $\frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}}$

   (ii) $ydx + \left(1+x^2\right)\tan^{-1}x\,dy = 0$

   (iii) $\sin\frac{dy}{dx} = a,\ y(0)=1$

   (iv) $\frac{dy}{dx} = e^{x+y} + x^3e^y$

Chapter 10 Differential Equation.indd 161 26-02-2020 17:24:47